

Control Statements

Control statements determine the flow of execution in a Java program. They decide which statements are executed, how many times they are executed, and under what conditions.

Java control statements are classified into three types:

1. Selection (Decision-Making) Statements
2. Iteration (Looping) Statements
3. Jump (Branching) Statements

Selection (Decision-Making) Statements

These statements execute different blocks of code based on a condition.

if Statement

```
public class Main {  
    public static void main(String[] args) {  
        int age = 20;  
  
        if(age >= 18) {  
            System.out.println("Eligible to vote");  
        }  
    }  
}
```

Eligible to vote

If-else Statement

```
public class Main {  
    public static void main(String[] args) {  
  
        int number = 15;  
  
        if(number % 2 == 0)  
            System.out.println("Even");  
        else  
            System.out.println("Odd");  
    }  
}
```

Odd

c) else-if Ladder

```
public class Main {
    public static void main(String[] args) {

        int marks = 82;

        if(marks >= 90)
            System.out.println("Grade A");
        else if(marks >= 75)
            System.out.println("Grade B");
        else if(marks >= 50)
            System.out.println("Grade C");
        else
            System.out.println("Fail");
    }
}
```

Grade B

Nested if

```
public class Main {
    public static void main(String[] args) {

        int age = 22;
        boolean hasID = true;

        if(age >= 18) {
            if(hasID)
                System.out.println("Entry Allowed");
        }
    }
}
```

Entry Allowed

Switch Statement

```
public class Main {
    public static void main(String[] args) {

        int day = 3;

        switch(day)
        {
            case 1:
                System.out.println("Monday");
                break;
        }
    }
}
```

```
case 2:
    System.out.println("Tuesday");
    break;

case 3:
    System.out.println("Wednesday");
    break;

default:
    System.out.println("Invalid Day");
}
}
}

Wednesday
```

Switch Fall-Through Example

```
public class Main {
    public static void main(String[] args) {

        int n = 2;

        switch(n)
        {
            case 1:
                System.out.println("One");

            case 2:
                System.out.println("Two");

            case 3:
                System.out.println("Three");

            default:
                System.out.println("Done");
        }
    }
}

Two
Three
Done
```

Iteration (Looping) Statements

While Loop

```
public class Main {
    public static void main(String[] args) {

        int i = 1;

        while(i <= 5)
        {
            System.out.println(i);
            i++;
        }
    }
}
```

1
2
3
4
5

Do-while Loop

```
public class Main {
    public static void main(String[] args) {

        int i = 1;

        do
        {
            System.out.println(i);
            i++;
        }
        while(i <= 5);
    }
}
```

1
2
3
4
5

For Loop

```
public class Main {
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++)
        {
```

```
        System.out.println(i);
    }
}

1
2
3
4
5
```

Enhanced for Loop (for-each)

```
public class Main {
    public static void main(String[] args) {

        int numbers[] = {10,20,30,40};

        for(int num : numbers)
        {
            System.out.println(num);
        }
    }
}

10
20
30
40
```

Jump (Branching) Statements

Break Statement

```
public class Main {
    public static void main(String[] args) {

        for(int i=1;i<=10;i++)
        {
            if(i==6)
                break;

            System.out.println(i);
        }
    }
}

1
```

2
3
4
5

Continue Statement

```
public class Main {  
    public static void main(String[] args) {  
        for(int i=1;i<=5;i++)  
        {  
            if(i==3)  
                continue;  
            System.out.println(i);  
        }  
    }  
}
```

1
2
4
5

Return Statement

```
public class Main {  
    static void display()  
    {  
        System.out.println("Hello");  
        return;  
    }  
    public static void main(String[] args) {  
        display();  
        System.out.println("Java");  
    }  
}
```

Hello
Java